

Chapter 5

Universal Arrows

5.1 Universal Arrows

Definition 5.1.1 (Universal arrows¹). Given any functor $\mathbf{F} : \mathcal{X} \longrightarrow \mathcal{A}$ and object A of $\mathcal{A}$, a *universal arrow from $\mathbf{F}$ to A* consists of an object X of $\mathcal{X}$ and an arrow $\varepsilon_A : \mathbf{F}X \longrightarrow A$ such that given any object Y of $\mathcal{X}$ and arrow $f : \mathbf{F}Y \longrightarrow A$ there is a unique arrow $g : Y \longrightarrow X$ of $\mathcal{X}$ that makes the following triangle commute:

$$\begin{array}{ccc}
 X & & \mathbf{F}X \xrightarrow{\varepsilon_A} A \\
 \uparrow g & & \uparrow \mathbf{F}g \\
 Y & & \mathbf{F}Y \xrightarrow{f} A
 \end{array}$$

Dually, a *universal arrow from A to $\mathbf{F}$* consists of an object X of $\mathcal{X}$ and an arrow $\eta_A : A \longrightarrow \mathbf{F}X$ such that given any object Y of $\mathcal{X}$ and arrow $f : A \longrightarrow \mathbf{F}Y$ there is a unique arrow $g : X \longrightarrow Y$ of $\mathcal{X}$ that makes the following triangle commute:

$$\begin{array}{ccc}
 A & \xrightarrow{\eta_A} & \mathbf{F}X \\
 & \searrow f & \downarrow \mathbf{F}g \\
 & & \mathbf{F}Y
 \end{array}
 \qquad
 \begin{array}{c}
 X \\
 \vdots g \\
 Y
 \end{array}$$

Example 5.1.2. For any objects A and B of a category $\mathcal{C}$ with all finite products, the pair B^A and $ev_{A,B}$ is universal from the functor $_ \times A$ to B . In this case, the adjuncts are called transposes.

Theorem 5.1.3. ²If X, ε_A is universal from $\mathbf{F}$ to A and there is an isomorphism $i : Y \rightarrow X$ then $Y, \varepsilon_A \circ \mathbf{F}i$ is also universal from $\mathbf{F}$ to A . Conversely, if X, ε_A is universal from $\mathbf{F}$ to A and Y, ε'_A is too, and then the unique arrow $i : Y \rightarrow X$ with $\varepsilon'_A = \varepsilon_A \circ \mathbf{F}i$ is iso.

Exercise 5.1.4. Prove Theorem 5.1.3.

5.2 Limits/Colimits as Universal Arrows

Definition 5.2.1 (Diagram of Shape $\mathbb{I}$). A *diagrams of shape* $\mathbb{I}$ are functors $\mathbb{I} \rightarrow \mathcal{C}$ in which $\mathbb{I}$ is small.

$\mathbb{I}$ is called an *index category*, which is often finite.

Definition 5.2.2 (Constant Diagram³). For an object A of $\mathcal{C}$, the *constant diagram* on A is the constant functor $\Delta A : \mathbb{I} \rightarrow \mathcal{C}$ which sends all objects of $\mathbb{I}$ to A and all morphisms of $\mathbb{I}$ to id_A .

Definition 5.2.3 (Diagonal Functor). The *diagonal functor* $\Delta : \mathcal{C} \rightarrow \mathcal{C}^{\mathbb{I}}$ is the functor which sends an object A of $\mathcal{C}$ to the constant diagram ΔA on A and a morphism $f : A \rightarrow B$ of $\mathcal{C}$ to the natural transformation $\Delta f : \Delta A \rightarrow \Delta B$ with components $(\Delta f)_I \stackrel{def}{=} f$ where I is an object of $\mathbb{I}$.

¹See McLarty (1992, p.88).

²See McLarty (1992, p.88, Theorem 10.1).

³Crole (1993, p.91)

Definition 5.2.4 (Limits and Colimits).

- A *limit* for a diagram $D : \mathbb{I} \longrightarrow \mathcal{C}$ is a universal arrow $\langle r, v \rangle$ from Δ to D . It consists of an object $r \in ob\mathcal{C}$, represented as $r = \varprojlim D$ or $Lim D$, and a natural transformation $v : \Delta r \dashrightarrow D$, which is universal among natural transformations $\tau : \Delta c \dashrightarrow D$ for objects $c \in ob\mathcal{C}$.
- A *colimit* for a diagram D is a universal arrow $\langle r, u \rangle$ from D to Δ . It consists of an object $r \in ob\mathcal{C}$, represented as $r = \varinjlim D$ or $Colim D$, and natural transformation $u : D \dashrightarrow \Delta r$ which is universal among natural transformations $\tau : D \dashrightarrow \Delta c$ for objects $c \in ob\mathcal{C}$.

Example 5.2.5. Let $\mathbb{I}$ be the empty category. Then the limit and the colimit for shape $\mathbb{I}$ are a terminal and initial objects.

Example 5.2.6. Let $\mathbb{I}$ be the small discrete category (namely, a category for which the only morphisms are identities). Then the limit and the colimit of shape $\mathbb{I}$ amount to products and coproducts.

Example 5.2.7. Let $\mathbb{I}$ be the category

$$0 \rightrightarrows 1$$

which has just two objects and two parallel non-identity morphisms. The limit and colimit are equalizer and coequalizer.

Example 5.2.8. Let $\mathbb{I}$ be the category:

$$\begin{array}{ccc} & & 1 \\ & & \downarrow \\ 0 & \longrightarrow & 2 \end{array}$$

The limit and colimit for the diagram are a pullback and a pushout.

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